Tsz Fei Wang

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Education

University of Waterloo | Honours Computer Science (Co-op) — Sept 2026 – May 2031 (expected)

Skills & Technologies

Languages: Python, Java, C, C++, TypeScript, JavaScript, HTML/CSS, SQL, R, GLSL
Frameworks & Libraries: React, Next.js, Node.js, FastAPI, Three.js, PyTorch, NumPy, Pandas, Qt/QML
Tools & Platforms: Git, Linux, Docker, GitHub Actions, Firebase, Elasticsearch, Vercel, Cloudflare Workers, Postman, Jupyter, KiCad

Experience

Century Frontier Asset Management — Jul 2026 – Aug 2026
Quantitative Research Intern — *Shenzhen, CN*

- Designed a point-in-time backtesting pipeline in **Python** with **pandas** and yfinance that only shows each model the data available on its prediction date, removing lookahead bias from LLM-driven equity research.
- Built a model-agnostic evaluation framework benchmarking stock-preference and inter-stock-relationship predictions from **9 LLM architectures** across **104 tickers**.
- Scored signal quality separately from each model's stated confidence, showing that confidence was a poor guide to whether a prediction held up.

Krembil Research Institute, World Community Grid — Mar 2026 – Jun 2026
Software Engineer Intern — *Toronto, ON*

- Designed and built the full-stack data access portal for Catrin, a bioinformatics database of **~4 million** transcription factor to target gene (TF-TG) associations across **33,000 genes**.
- Built most of the portal's frontend: the search, filter and results views researchers use to look up regulatory interactions by gene or transcription factor.
- Reconciled the schemas of **9 independent data sources** that describe the same interactions differently, so researchers can search and filter them as one dataset.

Mackenzie Cybersecurity Team — Jun 2025 – Jun 2026
Team Captain, President — *Toronto, ON*

- Captained the team to **First in Canada** in CyberTitan, Canada's largest cyberdefense competition, securing **Windows** and **Linux** systems and **Cisco** network configurations under time limits.
- Designed and delivered cybersecurity curriculum and class presentations that reached 60+ students.

Non-Trivial Research Fellowship — Mar 2025 – Aug 2025
Machine Learning Research Fellow — *Remote*

- Trained and evaluated a **PyTorch** audio deepfake classifier in **Python** on Kaggle that distinguishes real speech from AI-synthesized speech.

Projects

Technova Hackathon Badge | *KiCad, Hardware Lead* — *Oct 2026*

- Led hardware design of the attendee badge for ~300 hackers in **KiCad**: a **RISC-V** (CH32V203) board with an NFC tag for tap-to-phone check-in, an OLED screen, a D-pad, an addressable LED ring and USB-C LiPo charging.
- Routed the double-sided board (1,300+ tracks, 100+ vias) to a clean DRC, keeping copper out of the 42 mm NFC antenna coil, and added a boot jumper so the badge can be flashed over USB.
- Prepared the JLCPCB assembly package (gerbers, LCSC-matched BOM, rotation-corrected placement file), replacing out-of-stock parts and bringing the cost to about $7 per badge.

Submerge | *GLSL, QML, Python* — *Oct 2026 – Present*

- Developed an animated deep-ocean koi theme for KDE Plasma 5 and 6 (wallpaper, global theme, panel style, login screen) in **GLSL** and **QML**, with ~1,290 downloads on the KDE Store.
- Implemented a water ripple simulation in the shaders and a live system HUD showing CPU, GPU, memory, network, battery and the top process.
- Added power-aware frame pacing to keep the animation light on battery, and resolved blank rendering on NVIDIA/Wayland after a user report by porting the shaders to GLSL ES.

Physical Studio | *Python, React, Three.js, Docker* | *Winner, Hack the Valley 11* — *Oct 2026*

- Engineered the **Python** MCP server that lets an LLM design a project from a library of measured parts (dimensions, mounting holes, pins) and check its own design for collisions and clearance, built in a team of three.
- Implemented GLB scene export to a **React Three Fiber** viewer that animates each assembly step, and deployed the app with **Docker Compose** and Caddy on a Vultr server.

Awards & Honors

Gold & Silver Medalist, IT Systems Networking, Skills Ontario (Regional & Provincial) — Mar 2026 – May 2026
National Champion x2, 1st in Canada, Cybersecurity, $1,000 scholarship — May 2025, Mar 2026
National Finalist, 2nd in Canada, Cybersecurity, $1,600 scholarship — Mar 2025
1st Place, Canada, Mirai Project, Technology Innovation in Healthcare, $400 award — Jan 2025